

ENOCH J. KIM

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EDUCATION

University of California Irvine

Bachelor of Science in Aerospace Engineering GPA: 3.91

Irvine, CA

Sept. 2023 – March 2028

PROFESSIONAL WORK EXPERIENCE

Shield AI

Mechanical Hardware Test Engineering Intern

Frisco, TX

July 2026 – December 2026

- Owned end-to-end mechanical design and implementation of hardware-in-the-loop (HIL) test hardware; including, hand calculations, FEA, trade studies, margins-of-safety, GD&T drawings, and documentation; presented designs through reviews.
- Designed and fabricated machined universal PCBA fixture, eliminating custom fixtures for every PCB; saving time and cost.
- Assisted with the assembly of a fuel-tank pump test stand, supporting hardware integration and test operations.
- Rapidly executed a broad range of engineering tasks under tight timelines, including single-day turnaround efforts across mechanical design, assembly, and hardware integration; utilized *Siemens NX & Teamcenter PLM*, *Atlassian Jira*, and *Confluence*.

NASA Langley Research Center

Aircraft eVTOL Sustainable Composites Manufacturing Intern/Researcher

Hampton, VA

January 2025 – May 2025

- Conducted extensive research; published in a NASA Technical Memorandum paper on sustainable composite materials [link](#)
- Performed mechanical, thermal, and spectroscopic characterization of natural fibers, carbon fibers, and biodegradable polymers.
- Developed procedure for tensile testing individual natural fibers. Tested 300+ samples of fibers ranging 7–25 μm in diameter.
- Processed and segmented images in ImageJ and Dragonfly, analyzed data using MATLAB to characterize fiber cross-section
- Operated advanced equipment for composite development including, Thermogravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC), Tensile Testing Machine, Fourier Transform Infrared Spectroscopy (FTIR), Raman Spectroscopy, X-Ray Diffraction (XRD), Optical Microscopy, and High-Resolution Scanning Electron Microscopy (HR-SEM).

ENGINEERING PROJECT EXPERIENCE

University of California Irvine Rocket Project Liquids - LNG & LOX

Chief Engineer

Irvine, CA

Mar 2026 – Present

- Held single final technical authority for all engineering decisions on a student-designed, built, and tested METHALOX rocket.
- Oversaw development of new injector with propulsion leads, which eliminated interpropellant seals(suspected cause of past engine failure), switched to O-ring boss fittings (ORB), and utilized empirically derived discharge coefficient (Cd) for orifice sizing.
- Owned vehicle-level architecture, mass allocation, & trade studies impacting performance, stability, GD&T, and testing.
- Managed cross-functional engineering through design reviews, hardware integration, test campaigns, and vehicle verification.

Aerodynamics and Structures Vehicle Engineer

May 2025 – Mar 2026

- Performed static calculations to design the aero structure in *Siemens NX & Teamcenter*, considering strength, weight, and DFM.
- Created K-bottle manifold structure connecting four bottles; enabled uninterrupted 3,700 psi fill, reducing testing time by 2 hrs.
- Designed and fabricated a 5ft-tall, 8in-diameter solid rocket for recovery testing, successfully validating recovery systems.
- Owned the design and manufacturing of a carbon-epoxy composite fin can using wet layups and prepreg, reduced weight by 15%
- Designed the engine 1500 lbf thrust load cell mount to ensure accurate measurements, survived 2 hotfires without failure.
- Ran FEA of prepreg carbon fiber fin can in *Ansys Mechanical*, utilizing ACP to model anisotropic composite materials.
- Performed aerodynamic and aeroelastic analysis using *XFLR5* and *Ansys Fluent* to optimize fin geometry, reduced drag by 10%

FSAE Anteater Formula Racing

Aerodynamics Design, Simulation, and Manufacturing Engineer

Irvine, CA

July 2024 – July 2025

- Increased front wing downforce by ~25% compared to the previous car by utilizing multi-element high-lift wing configuration .
- Optimized wing tip vortices to reduce dirty wake from tires and channel correct mass flow rate of air to radiator .
- Manufactured carbon fiber airfoils, body panels, and wings; created molds with CNC machining and tooling foam.
- Simulated and evaluated airflow, downforce, and drag using CFD in *Ansys Fluent* and quickly compared airfoils in *XFLR5*.
- Utilized MATLAB and simulations at different AoA, Reynolds number, and size to identify optimal wing configuration

SKILLS & INTERESTS

Software & Coding Languages: Siemens NX & Teamcenter, SolidWorks, Ansys Fluent & Mechanical, Onshape, TinkerCAD, C++, Python, Java, MATLAB, Arduino, Microsoft Excel & Office Suite, Adobe Photoshop, After Effects, InDesign

Hardware: Composite Manufacturing, Machining, Mig Welding, Soldering, 3D printing, Laser Cutter, TGA, DSC, FTIR, RAMAN, HRSEM, Tensile Testing, XRD, OM

Interests: Rockets, Jets, eVTOLs/AAMs, FPV Drone flying, Snowboarding, Basketball, Video Editing & Photography